Section XVI

OVERDOSES & POISONS

Poisons and medicine are oftentimes the same substance given with different intents.

Peter Latham (1865)

Chapter 52

Pharmaceutical Drug Overdoses

A desire to take medicine is, perhaps, the great feature which distinguishes man from other animals.

Sir William Osler (*a*)

Prescription drug use in the United States is commonplace, (see Figure 52.1), indicating that man's desire to take drugs may be matched by physicians' desire to prescribe them, and this may explain, at least in part, why deaths from drug overdoses have increased more than 5-fold in the past 20-odd years (from 20,000 deaths in the year 2000, to 108,000 deaths in 2022) (1). This chapter describes some of the more frequent or more problematic pharmaceutical drug overdoses, including intoxications from acetaminophen, benzodiazepines, β-blockers, opioids, and salicylates. You won't have to spend much time in an ICU to encounter one of these overdoses.

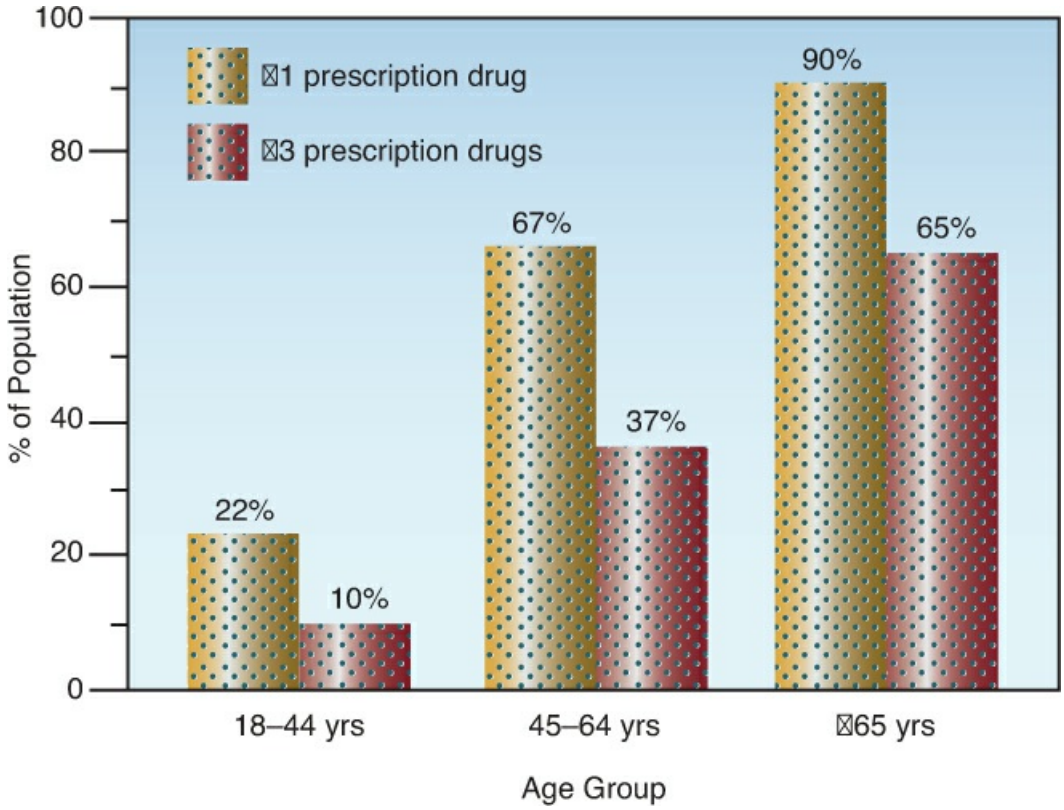


FIGURE 52.1 Prescription drug use in the United States, 2015–2018, by age group. Data from the National Center for Health Statistics, Health United States, 2019: Table 039. Hyattsville, MD, 2019. Available from www.cdc.gov/nchs/hus/data-finder.htm (Accessed 7/16/2024).

ACETAMINOPHEN

Acetaminophen is a ubiquitous, over-the-counter analgesic-antipyretic agent that is included in over 600 commercial drug preparations. It is also a hepatotoxin, and is *the leading cause of acute liver failure in North America, Europe, and Australia* (2). In the United States, acetaminophen is responsible for almost half of the cases of acute liver failure (see Figure 39.1) (3), and about *half of the overdoses are unintentional* (4). In 2011, the U.S. Food and Drug Administration (FDA) issued a mandate for more prominent labeling of the risks of acetaminophen hepatotoxicity, along with a requirement that prescription acetaminophen products be limited to 325 mg per dosage (5). Unfortunately, this mandate has had little impact on the prevalence of acute liver failure from acetaminophen. (See A FINAL WORD for more on this issue.)

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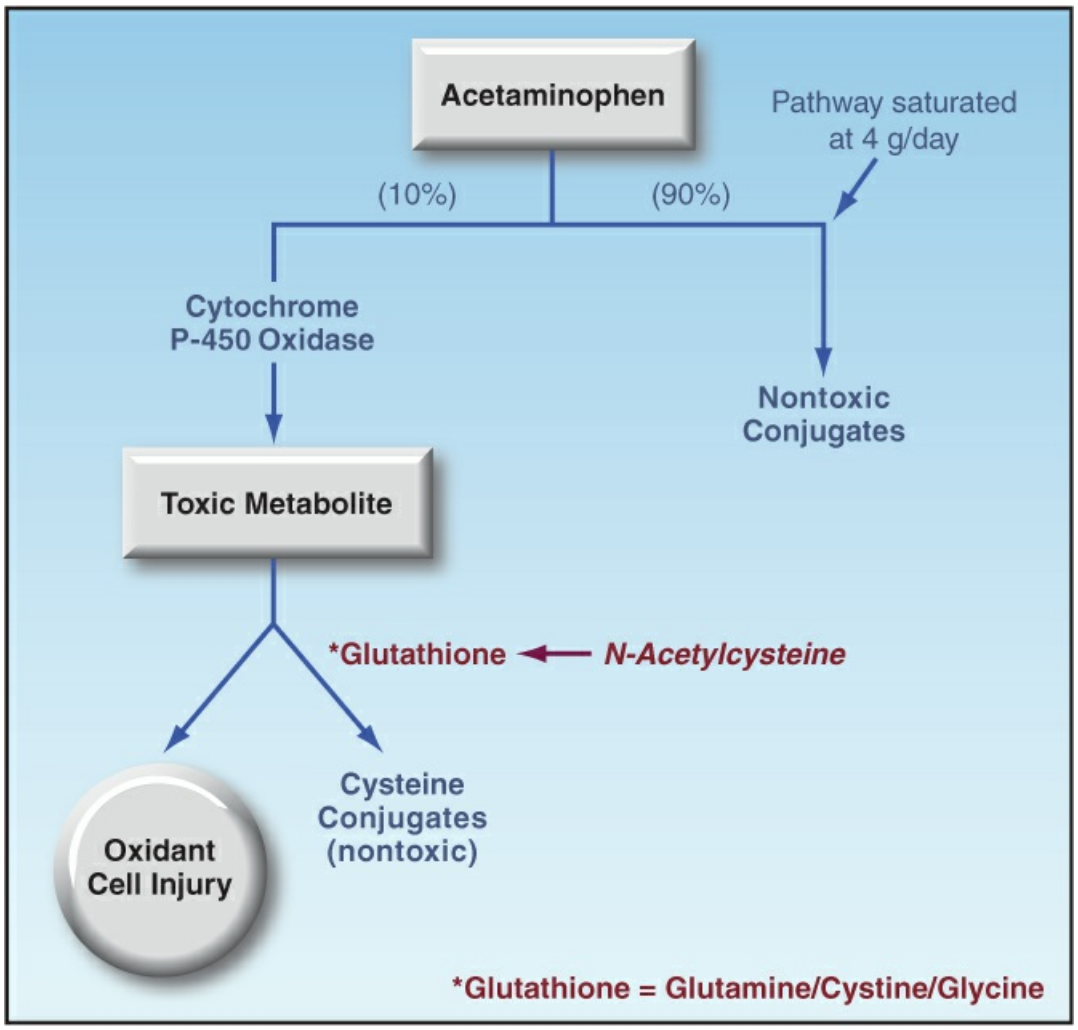


FIGURE 52.2 The hepatic metabolism of acetaminophen. See text for explanation.

Toxic Mechanism

The toxicity of acetaminophen is related to its metabolism in the liver, and the metabolic pathways are shown Figure 52.2. The bulk of acetaminophen metabolism involves the formation of nontoxic sulfate and glucuronide conjugates, which are excreted in the urine. About 10% of the metabolism involves the oxidation of acetaminophen by a cytochrome P450 oxidase group of enzymes to form a toxic metabolite that is capable of lethal cell injury. When the daily dose of acetaminophen is not excessive, the toxic metabolite is removed by conjugation with the cysteine moiety in glutathione, the major intracellular antioxidant. (*Note:* Glutathione is a tripeptide that is composed of cysteine. glutamine, and glycine.) When the daily dose of acetaminophen is excessive (i.e., above 4 grams/day), the nontoxic conjugation pathway becomes saturated, which diverts acetaminophen metabolism to the pathway with the toxic metabolite. The increased traffic in this pathway begins to deplete glutathione reserves, and when glutathione levels fall by 70%, the toxic acetaminophen metabolite accumulates and causes hepatocellular injury (3).

Toxic Dose

. The toxic dose of acetaminophen can vary in individual patients, but is somewhere between 7.5 and 15 grams daily in most adults (6,7). Many poison control centers use 10 grams as a threshold dose that requires further investigation (8). However, certain conditions can increase the susceptibility to acetaminophen hepatotoxicity (see next section).

. According to the FDA, the maximum dose of acetaminophen that is safe in adults is 4 grams per day (9). However, some clinical practice guidelines recommend that for chronic acetaminophen use, the daily dose should not exceed 3 grams (10).

Predisposing Conditions

The following conditions can increase the risk of liver damage from acetaminophen. In the presence of these conditions, a daily acetaminophen dose of >4 grams should be considered as potentially toxic (8).

. Malnutrition, or prolonged periods of fasting, are recognized risk factors because of the potential for depletion of hepatic glutathione stores.

. Patients with chronic alcohol use disorder are at increased risk of acetaminophen hepatotoxicity, *even when daily acetaminophen doses are below the recommended safe level (i.e., <4 grams daily)* (11). This is likely due to the ability of alcohol to increase the activity of the cytochrome P450 enzyme, plus the possible contribution of glutathione depletion from poor nutrition.

. Patients with chronic liver disease are also considered high risk for acetaminophen hepatotoxicity (8), although the evidence for this is not convincing (12).

. Drugs that stimulate cytochrome P450 activity can increase the risk of liver damage from acetaminophen. These drugs include anticonvulsants (carbamazepine and phenytoin) and antituberculosis drugs (INH and rifampin) (12). Opioids can also increase the risk by competing for the nontoxic conjugation pathway, thereby diverting acetaminophen metabolism toward the production of the toxic metabolite (12).

Clinical Presentation

The clinical presentation of acetaminophen hepatotoxicity can be divided into three stages:

. *Stage 1 (First 24 hours):* Symptoms are absent or nonspecific (e.g., nausea, vomiting, malaise), and there may be no evidence of hepatic injury.

. *Stage 2 (24–72 hours):* Hepatic aminotransferases (aspartate aminotransferase [AST], alanine aminotransferase, [ALT]) begin to rise at 24 hours, and can reach levels of 10,000 IU/L (13). Signs of progressive hepatic injury (e.g., jaundice and prolonged INR) begin to appear, and there may be signs of renal impairment.

. *Stage 3 (72–96 hours):* The hepatic injury peaks at 72–96 hours (12). Hepatic encephalopathy may be evident at this time, along with progressive lactic acidosis and acute, oliguric renal failure. Death from multiorgan failure usually occurs at this time.

Risk Assessment

The evaluation of acetaminophen overdoses must include the following: (*a*) the time and dose

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ingested, (*b*) the type of ingestion (acute overdose versus unintentional supratherapeutic dosing), (*c*) the drug preparation (immediate vs. extended release), (*d*) the presence of predisposing conditions, and (*e*) evidence of hepatic injury. If an overdose presents within 24 hours of ingestion of an immediate-release preparation, the plasma acetaminophen level can identify the need for antidotal therapy.

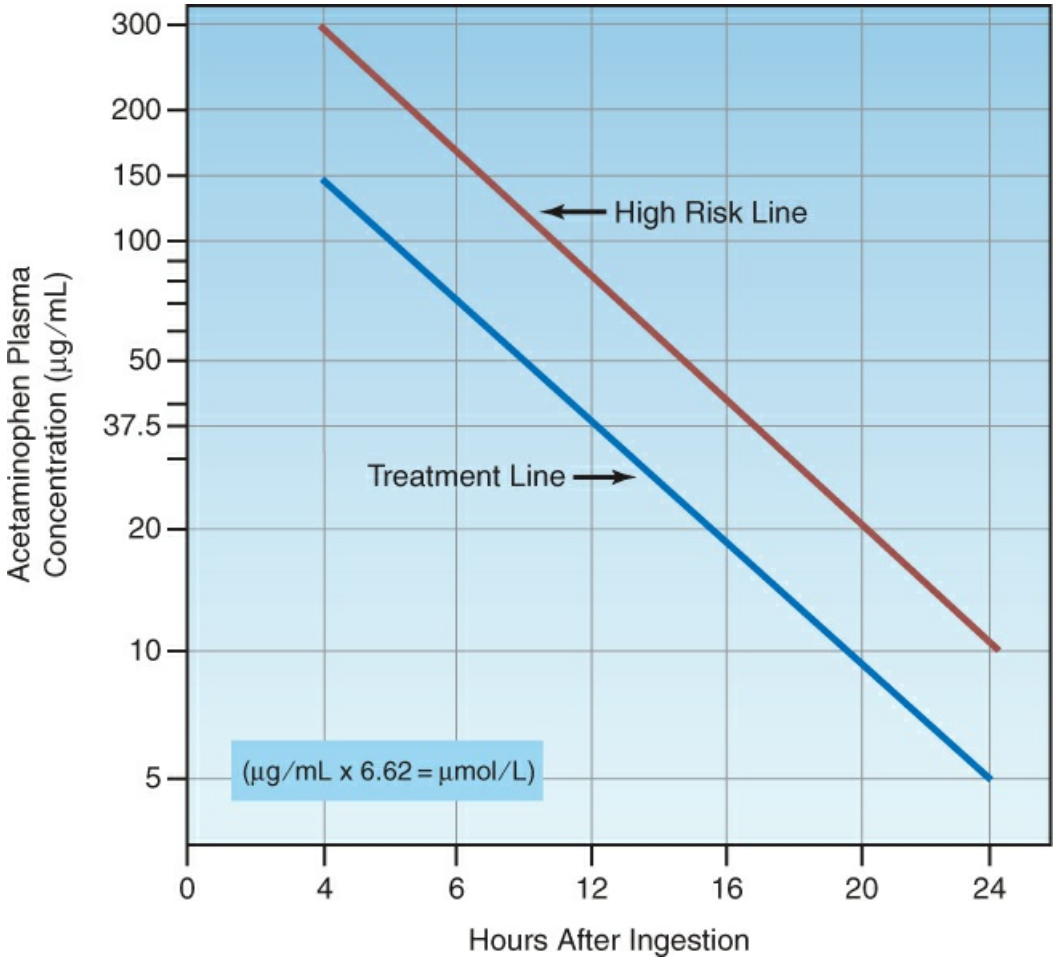


FIGURE 52.3 Nomogram for predicting the risk of hepatotoxicity according to the plasma acetaminophen level between 4 and 24 hours after ingestion. See text for explanation. Nomogram redrawn from Reference 14.

Predictive Nomogram

Plasma acetaminophen levels obtained from 4 to 24 hours after an acute drug ingestion can be used to predict the risk of hepatotoxicity using the nomogram in Figure 52.3 (14). If the plasma level is above the treatment line (which extends from 150 µg/mL at 4 hrs, to 5 µg/mL at 24 hrs), the risk of hepatotoxicity is high enough to warrant antidotal therapy with N-acetylcysteine. If the plasma level is above the high-risk line (from 300 µg/mL at 4 hrs, to 10 µg/mL at 24 hrs), there is a 90% chance of developing hepatotoxicity, and these cases may require extended treatment with N-acetylcysteine (see later).

The predictive nomogram is useful only for acute ingestions that can be accurately timed. If an extended-release preparation was ingested and the plasma acetaminophen level is not above

the treatment line at 4–12 hours after ingestion, a repeat level should be obtained after another 4–6 hours (14). For repeated supratherapeutic ingestions, plasma levels do not predict the risk of hepatotoxicity, and evaluating risk can be challenging.

N-Acetylcysteine

Glutathione does not readily cross cell membranes, so exogenous glutathione is not a viable treatment option for acetaminophen hepatotoxicity. However, glutathione is a tripeptide (i.e., glutamine, cysteine, and glycine), and the cysteine moiety is involved in clearing the toxic acetaminophen metabolite, so a cysteine surrogate, such as N-acetylcysteine (NAC), is a logical (and effective) antidote for acetaminophen hepatotoxicity (see Fig. 52.2).

Treatment with NAC is indicated for any patient with a plasma acetaminophen level above the treatment line in the predictive nomogram. However, *NAC is also recommended for any case of potential acetaminophen hepatotoxicity when the hepatic aminotransferase levels are increased* (14).

Intravenous Regimen

The treatment with NAC can be intravenous or oral. The intravenous route is preferred, and the IV regimen is shown in Table 52.1. The treatment proceeds in two stages (14). The first stage involves a 21-hour infusion of NAC, which is given in 3 separate aliquots, at a total dose of 300 mg/kg (2). This is followed by an evaluation of 4 goals of treatment, which are listed in Table 52.1 as the "NAC stopping criteria". If these goals have not been achieved, the NAC infusion is continued at a dose rate of at least 6.25 mg/kg/hr (14), and the goals are re-evaluated again 12–24 hours later. The NAC infusion is continued until the goals have been achieved.

TABLE 52.1 Intravenous N-Acetylcysteine for Acetaminophen Hepatotoxicity

1. Regimen:

Use 20% NAC (200 mg/mL) for each of the doses below, and infuse in sequence.

1. 150 mg/kg in 200 mL D_5W over 1 hr.
2. 50 mg/kg in 500 mL D_5W over 4 hrs.
3. 100 mg/kg in 1,000 mL D_5W over 16 hrs.

Total Dose: 300 mg/kg over 21 hrs.

2. NAC Stopping Criteria:

1. Plasma acetaminophen <10 μg/mL.
2. AST/ALT normal for patient, or decreased by 25–50%.
3. INR <2.0.
4. Patient feeling well.

a. If NAC stopping criteria are not satisfied, continue IV NAC at a rate of at least 6.25 mg/kg/hr.
b. Recheck criteria every12–24 hrs, and continue NAC until criteria are satisfied.

NAC Regimen from Reference 2; NAC stopping criteria from Reference 14.

Oral Regimen

There is an oral NAC regimen that is as effective as the IV regimen (15), but it is rarely used

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because NAC has a very disagreeable taste (due to the sulfur content). The oral regimen is as follows:

. Use 10% NAC, and dilute 2:1 in water to make a 5% solution (50 mg/mL).
. Start with a dose of 140 mg/kg, and follow with a maintenance dose of 70 mg/kg every 4 hours for 72 hours.
. The total dose is 1,330 mg/kg, administered over 72 hours.

There are no clear instructions on checking for improvement or extending the treatment period for oral NAC, as there is for intravenous NAC.

Adverse Reactions

Adverse reactions to intravenous NAC include nausea, vomiting, and cutaneous hypersensitivity reactions (rash, angioedema), which are more common in the first hour (when the NAC dose is the largest), and have a reported incidence that varies from 9% to 77% (16). Other IV regimens have been proposed that reduce the frequency of adverse reactions (2,16), but the efficacy of these regimens is unclear.

Hemodialysis

The elimination of acetaminophen is enhanced by hemodialysis, which is recommended, in addition to NAC, when the plasma acetaminophen concentration exceeds 900 μg/mL, and there is evidence of hepatic encephalopathy (14). During hemodialysis, the intravenous NAC dose should be at least 12.5 mg/kg/hr (because NAC can also be eliminated by dialysis). No dose adjustment is required for the oral NAC regimen.

Outcomes

Most patients respond favorably to NAC, and the reported mortality rate in acetaminophen hepatotoxicity is 0.4% (2). However, an occasional patient will show progressive signs of hepatic injury despite NAC treatment, and these patients should be considered for liver transplantation.

BENZODIAZEPINES

Benzodiazepines are second only to opiates as the drugs most frequently involved in medication-related deaths (17). However, benzodiazepines are rarely fatal when ingested alone (18), and other respiratory depressant drugs (e.g., opiates) are almost always involved in benzodiazepine-related fatalities (17).

Clinical Features

Because overdoses involving benzodiazepines also involve other drugs, the clinical presentation can vary (according to the drugs ingested). Pure benzodiazepine overdoses produce deep sedation but rarely result in coma (18). Respiratory depression (2–12% of cases), bradycardia (1–2% of cases) and hypotension (5–7% of cases) are also uncommon (18). Benzodiazepine intoxication can also produce an agitated confusional state, with hallucinations, that could be

mistaken for alcohol withdrawal (18).

Involvement of benzodiazepines in an apparent overdose can be difficult to establish because there are no serum assays for benzodiazepines, and screening tests for benzodiazepines in urine can miss common agents like lorazepam (19). As a result, benzodiazepine involvement is usually based on the clinical history.

Management

The management of benzodiazepine overdose involves general supportive care, including blood pressure support and mechanical ventilation, if needed. An antidote (flumazenil) is available, but is not popular.

Flumazenil

Flumazenil is a benzodiazepine antagonist that binds to benzodiazepine receptors (GABA-A receptors), but has minimal agonist activity (20). It is effective in reversing benzodiazepine-induced sedation, but is inconsistent in reversing benzodiazepine-induced respiratory depression (21,22).

DOSING REGIMEN: Flumazenil is given as intravenous boluses of 0.2 mg that can be repeated every few minutes to a cumulative dose of 1.0 mg. The response is rapid, with onset in 1–2 minutes, and peak effect at 6–10 minutes (20). The effect lasts about one hour. Since flumazenil has a shorter duration of action than the benzodiazepines, sedation can return after 30–60 minutes. To reduce the risk of re-sedation, the initial dose of flumazenil can be followed by a continuous infusion at 0.3–0.4 mg/hr (23).

ADVERSE REACTIONS: Flumazenil has suffered from reports of frequent adverse effects, as shown in Table 52.2. The data in this table represents the pooled results of 13 clinical trials that compared flumazenil to placebo in patients who presented to an emergency department with impaired consciousness from suspected benzodiazepine overdose (23). As indicated, the minor and major adverse events were significantly more frequent with flumazenil, but the major events (which included seizures) were infrequent, and were significantly different from placebo only at the $p < 0.05$ level. *The issue that is not addressed in these studies is the benefit of awakening with flumazenil.*

TABLE 52.2 Relative Frequencies of Adverse Events with Flumazenil

	Flumazenil	Placebo	p value
Number of Patients	498	492	
Minor Adverse Events†	28%	10%	<0.001
Major Adverse Events§	2.4%	0.4%	0.02

†Most common minor events were agitation, nausea, and vomiting. §Most common major events were supraventricular arrhythmias and seizures. Data from Reference 23.

Because of the risks associated with flumazenil, it is not recommended for patients who present with coma of undetermined etiology, or for patients who have a history of seizures or

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chronic benzodiazepine dependence. Its major use at the present time is for rapid awakening following procedural sedation.

B-RECEPTOR ANTAGONISTS

Intentional β-blocker overdoses are uncommon, but can be life-threatening. An effective antidote is available, if needed.

Toxic Manifestations

The typical manifestations of β-blocker overdose are *bradycardia* and *hypotension* (24). The bradycardia is usually sinus in origin, and is well tolerated. The hypotension can be due to peripheral vasodilation (renin blockade), or a decrease in cardiac output (negative inotropic effect), or both.

Membrane Stabilizing Effect

Excessive doses of β-blockers can exert a membrane stabilizing effect that is independent of the β-receptor blockade, and can be life-threatening. The principal consequence is prolonged atrioventricular (AV) conduction, which can progress to complete heart block (25).

Neurotoxicity

Most β-blockers are lipophilic, and have a tendency to accumulate in lipid-rich tissues like the central nervous system, where they can promote *lethargy*, *depressed consciousness*, and *generalized seizures*. The latter manifestation is more prevalent than suspected, and has been reported in 60% of overdoses with propranolol (26). The CNS effects of β-blockers are the result of the membrane stabilizing effect, and not β-receptor blockade.

Glucagon

Glucagon is a regulatory hormone that acts in opposition to insulin by stimulating glycogen breakdown to raise blood glucose levels. In a seemingly unrelated role, glucagon antagonizes the cardiac depression produced by β-receptor antagonists.

Mechanism of Action

The illustration in Figure 52.4 shows how glucagon can correct the cardiac depression caused by β-blockers and calcium channel blockers. The glucagon receptor and β-receptor are linked to the adenyl cyclase enzyme on the inner surface of the cell membrane. Activation of each receptor-enzyme complex results in the hydrolysis of adenosine triphosphate (ATP) and the formation of cyclic adenosine monophosphate (cyclic AMP). The cyclic AMP then activates a protein kinase that promotes the inward movement of calcium through the cell membrane. The influx of calcium promotes interactions between contractile proteins to enhance the strength of cardiac contraction.

The sequence of reactions just described is responsible for the positive inotropic and chronotropic effects of β-receptor stimulation. Since the same reactions occur with activation of the glucagon receptor, *glucagon has cardiostimulatory effects that are not affected by β-receptor*

blockade (27). This allows glucagon to serve as an antidote for β-blocker overdoses.

Clinical Use

Glucagon is indicated for the treatment of hypotension and *symptomatic* bradycardia associated with toxic exposure to β-blockers. When used in the appropriate doses, glucagon will elicit a favorable response in 90% of patients (27). However, *glucagon will not reverse the prolonged AV conduction or neurological abnormalities in β-blocker overdoses* because these effects are not mediated by β-receptor blockade.

CALCIUM ANTAGONIST TOXICITY: Glucagon is also capable of antagonizing the effects of calcium channel blockers, as illustrated in Figure 52.4. However, glucagon is less effective in reversing the cardiac depression in calcium channel blocker overdoses.

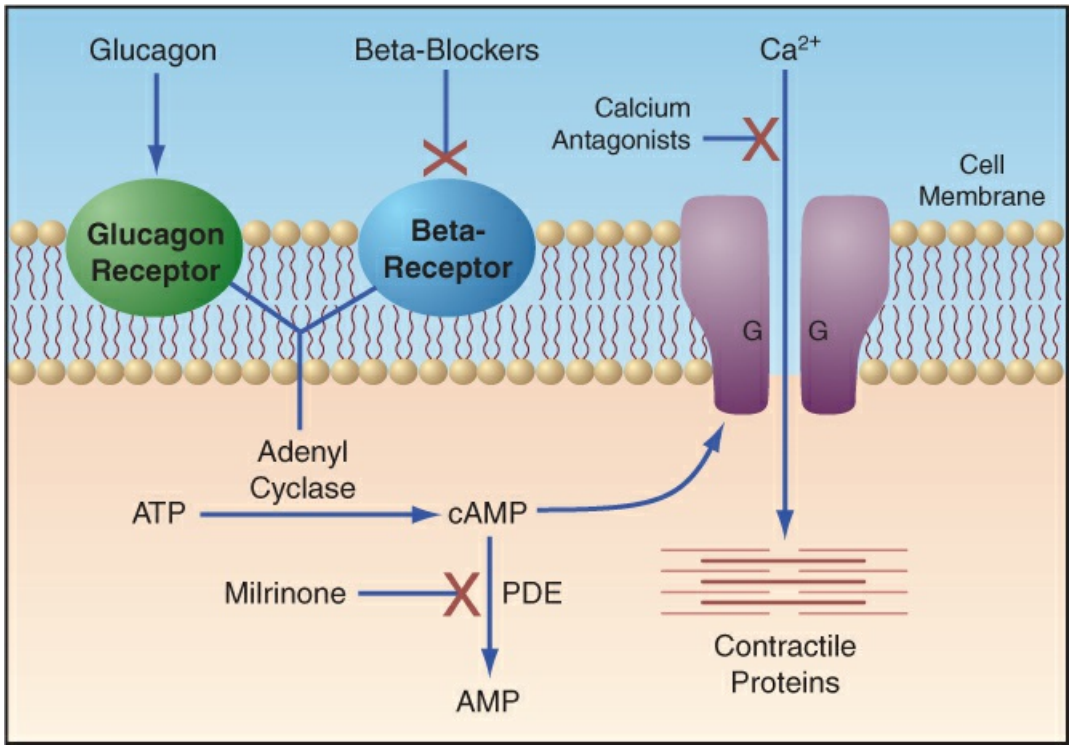


FIGURE 52.4 Mechanisms of drug-induced alterations in cardiac contractile strength. See text for explanation.. Abbreviations: ATP = adenosine triphosphate, cAMP = cyclic adenosine monophosphate, PDE = phosphodiesterase, AMP= adenosine monophosphate.

Dosing Regimen

Dosing recommendations for glucagon are as follows:

. The effective dose of glucagon can vary in individual patients, but *a bolus dose of 3–5 mg IV should be effective in most adults* (26,28). The initial dose is usually 3 mg (or 0.05 mg/kg), and the response should be evident within 3 minutes (29).

. If the response to the initial dose is not satisfactory, a second IV bolus dose of 5 mg (or 0.07 mg/kg) can be given.

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. The chronotropic response to glucagon is optimal when the plasma ionized calcium is normal (30).

. The effects of glucagon can be short-lived (5 minutes), and so a favorable response should be followed by a continuous infusion (5 mg/hr).

Adverse Effects

Nausea and vomiting that are refractory to ondansetron are common at glucagon doses above 5 mg/hr (27). Mild hyperglycemia is common, and is the result of glucagon's actions to stimulate glycogenolysis. The insulin response to the hyperglycemia can drive potassium into cells and promote hypokalemia. Finally, glucagon stimulates catecholamine release from the adrenal medulla, and this can cause unwanted increases in blood pressure in patients with hypertension (27).

Phosphodiesterase Inhibitors

Phosphodiesterase inhibitors (e.g., milrinone) augment cardiac contractile strength by inhibiting the breakdown of cyclic AMP, as illustrated in Figure 52.4. These agents can increase cardiac output in the setting of β-blockade (31), and they should add to the increase in cyclic AMP produced by glucagon. However, it is unclear if phosphodiesterase inhibitors add to the effectiveness of glucagon in β-blocker toxicity. Because these drugs are vasodilators, and can produce unwanted decreases in blood pressure, they are generally reserved for the occasional case of β-blocker toxicity that does not respond to glucagon.

OPIOIDS

The problem with opioid overdoses in the United States is certainly no secret, and most cases involve prescription opioids rather than street drugs (32). The adverse effects of opioids are described in Chapter 6, and this section focuses on the use of the narcotic antagonist, naloxone.

Clinical Features

The classic "textbook" description of an opiate overdose is the patient with stupor, pinpoint pupils, and slow breathing (bradypnea), but in reality, it is not possible to identify an opiate overdose based on the clinical presentation or physical examination (33). The response to the narcotic antagonist, naloxone, is probably the most reliable method of identifying an opioid overdose.

Naloxone

Naloxone (Narcan®) is a pure opioid antagonist; i.e., it binds to endogenous opioid receptors, but does not elicit any opioid-like responses. It is most effective in blocking mu (μ) receptors, which are responsible for analgesia, euphoria, and respiratory depression (33). Naloxone can be given by intranasal spray or intravenous injection.

Intranasal Naloxone

The intranasal route is favored for naloxone administration because it is as effective as the

intravenous route (34), but is much easier to administer, and can be given in the field with minimal training.

- Narcan is available as a single-dose nasal spray containing 2 mg or 4 mg naloxone hydrochloride in 0.1 mL of diluent (35). A single dose is delivered into one nostril.
- The recommended dose is either 2 mg or 4 mg, but 2 mg is probably sufficient in most cases, since intranasal doses of 0.4 mg have been effective in clinical studies (34). If there is no response within 3 minutes after the initial dose, a second dose can be given in the other nostril. If the 4 mg dose of naloxone is used, it may be more effective if given in two divided (2 mg) doses in each nostril, as suggested by the plasma naloxone levels in Fig. 52.5 (35).
- The average response time to intranasal naloxone is about 3 minutes (34), and peak blood levels are reached after about 20 minutes (see Fig. 52.5). The elimination half life is between 1.5 and 2 hours (35).
- Intranasal naloxone is effective for reversing both the impaired consciousness and the respiratory depression caused by opioids (34,35). However, the opioid effects are likely to return after 1–2 hours, so a favorable response to naloxone should be followed by a continuous intravenous infusion, at an hourly dose that is two-thirds of the effective dose (diluted in 250 or 500 mL of isotonic saline and infused over 6 hours) (36).

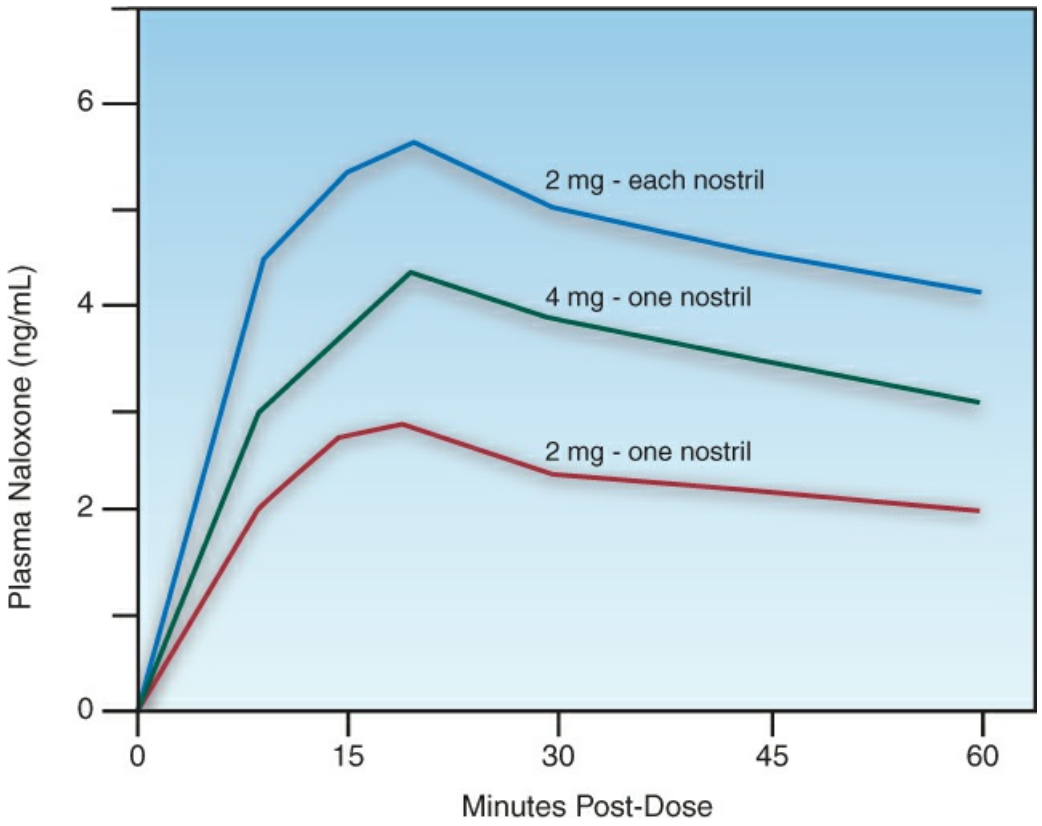


FIGURE 52.5 Plasma naloxone levels after intranasal naloxone in the doses indicated. Note that a dose of 4 mg results in higher plasma levels if the dose is divided, and 2 mg is administered in each nostril. Data from Reference 35.

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Intravenous Naloxone

Despite the advantages of intranasal naloxone, intravenous naloxone continues to be the gold standard method for treating opioid overdoses in the hospital setting. The IV route provides greater bioavailability, and a quicker response time, than the intranasal route (34,37) (although the clinical relevance of these differences is unproven). However, the IV route is generally reserved for situations where the nasal spray is not immediately available.

. For patients with a depressed sensorium but no respiratory depression, the initial dose of IV naloxone should be 0.4 mg. This can be repeated in 2 minutes, if necessary. A total dose of 0.8 mg should be effective if the mental status changes are caused by an opioid derivative (38).

. Respiratory depression can be more difficult to reverse than depressed consciousness. The initial naloxone dose for respiratory depression should be 2 mg IV push (33). If there is no response in 2–3 minutes, double the dose to 4 mg. Further doses, if necessary, can be increased to 15 mg. If there is no response to a dose of 15 mg, then opioids are not the cause of the respiratory depression (33).

. The reversal by IV naloxone lasts only about one hour, and opioid effects will return thereafter. Therefore, a favorable response to IV naloxone should be followed by a continuous infusion, using the dosing recommendations described for intranasal naloxone.

Adverse Reactions

Naloxone has few undesirable effects. The most common adverse effect is opioid withdrawal (39). There are also case reports of acute pulmonary edema associated with high-dose naloxone, which is attributed to pulmonary venoconstriction from activation of the sympathetic nervous system (39).

SALICYLATES

Salicylate poisoning receives relatively little attention, but it is second only to acetaminophen poisoning as a cause of suicide-related deaths from over-the-counter analgesics (40). Ingestion of 10–30 grams, or 150 mg/kg, of acetylsalicylic acid (aspirin) can have fatal consequences.

Pathogenesis

Once ingested, acetylsalicylic acid is converted to *salicylic acid*, which is the active form of the drug. Salicylic acid is readily absorbed from the upper GI tract, and metabolism takes place in the liver. *The hallmark of salicylate intoxication is the combination of a respiratory alkalosis and a metabolic acidosis with an elevated anion gap* (41).

Respiratory Alkalosis

Salicylic acid directly stimulates the respiratory centers in the lower brainstem, producing an increase in respiratory rate and tidal volume, and a subsequent decrease in the arterial PCO_2 (i.e., acute respiratory alkalosis). This is one of the earliest signs of salicylate poisoning.

Metabolic Acidosis

Salicylic acid is a weak acid that does not readily dissociate, and thus does not produce a metabolic acidosis. However, salicylic acid uncouples oxidative phosphorylation, which results in a compensatory increase in glycolysis (to generate ATP) and an increase in lactate production. The accumulation of lactic acid (and also ketoacids) results in a high anion-gap metabolic acidosis. This is a relatively late complication, and an acidemic pH is a poor prognostic sign (42).

Other Features

Salicylate-induced uncoupling of oxidative phosphorylation results in an increased metabolic rate, which increases the body temperature and promotes diaphoresis and dehydration.

Clinical Manifestations

The earliest signs of salicylate intoxication are hyperpnea, tinnitus, vertigo, nausea and vomiting, which can appear in the first few hours after a toxic ingestion. These can be followed by agitation and fever, and in severe cases, by coma, cerebral edema, acute respiratory distress syndrome (ARDS), and hemodynamic instability (41). Laboratory studies can reveal hypocapnia and respiratory alkalosis, lactic acidosis, rhabdomyolysis, hypernatremia (from dehydration) and acute kidney injury.

Diagnosis

The plasma salicylate level is used to confirm or exclude the diagnosis of salicylate toxicity.

. The therapeutic plasma salicylate level is 10–30 mg/dL (0.7–2.2 mmol/L).

. Plasma salicylate levels above 40 mg/dL (2.9 mmol/L) are considered toxic, and levels above 75 mg/dL (5.4 mmol/L) are considered life-threatening (41).

. Salicylate levels are often lower in cases of chronic salicylate toxicity.

Management

Management of salicylate toxicity includes general supportive care (e.g., fluids, vasopressors, and mechanical ventilation), plus the following measures.

Activated Charcoal

Because salicylates can slow gastric emptying, and enteric-coated or extended-release aspirin preparations can promote retention of the drug in the gastric lumen, activated charcoal is recommended for all salicylate intoxications, regardless of the time of drug ingestion (41). (However, activated charcoal is most effective if given within 2 hours of drug ingestion.) The dose is 1 g/kg to a maximum dose of 100 grams, which is repeated every 4 hours until the charcoal appears in the stool or the plasma salicylate levels begin to decrease (41).

Alkalinization of the Urine

Alkalinization of the urine increases salicylate excretion, and is one the cornerstones of management for salicylate intoxication. Bicarbonate infusions are used to raise the urine pH using the regimen in Table 52.3 (42,43). This should be continued until the plasma salicylate

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levels fall into the therapeutic range.

TABLE 52.3 Protocol for Alkalinization of the Urine

1. Create a bicarbonate solution by adding 3 amps $NaHCO_3$ (44 mEq /amp) to 1 liter D_5W (132 mEq/L). Add 40 mEq KCL.
2. Start with a bicarbonate loading dose of 1–2 mEq/kg.
3. Follow with infusion of the bicarbonate solution at 2–3 mL/kg/hr.
4. Maintain a urine output of 1–2 mL/kg/hr, and a urine pH ≥7.5.

From References 42,43.

HYPOKALEMIA: Bicarbonate infusions will lower the serum potassium (by driving K^+ into cells), and hypokalemia hampers the ability to alkalinize the urine (by increasing the secretion of H^+ into the lumen of the distal tubules). This provides a reason to add K^+ (40 mEq) to the bicarbonate solution, as shown in Table 52.3.

Hemodialysis

Hemodialysis is the most effective method of clearing salicylates from the body (41,44). The indications for hemodialysis include (41): (*a*) a plasma salicylate level >90 mg/dL (>6.5 mmol/L), (*b*) a salicylate level >80 mg/dL (>5.8 mmol/L) plus impaired renal function, or (*c*) evidence of life-threatening intoxication (e.g., cerebral edema, ARDS, multiorgan failure) (41).

A FINAL WORD

Regulating Acetaminophen

If any over-the-counter drug deserves to be regulated, it's acetaminophen. Not only because it is the leading cause of acute liver failure in North America, Europe, and Australia, but also because *half of the overdoses are unintentional* (i.e., the drug ingestion is for therapeutic reasons), indicating a general lack of awareness of the potential for harm with acetaminophen. The FDA issued a mandate for stronger warnings on drug packaging, but who reads that small print? Requiring a prescription that indicates how the drug should be used seems to be the right step for promoting the safe use of acetaminophen.

Acetaminophen gained its popularity in the 1970s because of concerns about the toxicity of aspirin, and it looks like we replaced a potentially toxic drug with a potentially lethal one.

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Acetaminophen

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Chapter 53

Nonpharmaceutical Poisons

It is astonishing with how little reading a doctor can practice medicine, but it is not astonishing how badly he may do it.

Sir William Osler (*a*)

This chapter describes the toxic syndromes produced by exposure to nonpharmaceutical poisons, including carbon monoxide, cyanide, and the toxic alcohols (methanol and ethylene glycol). They are less common than the drug overdoses presented in the last chapter, but they can be lethal if not recognized.

CARBON MONOXIDE

Carbon monoxide (CO) is produced by the incomplete combustion of organic (carbon-based) matter, and is one oxidation reaction shy of carbon dioxide ($2CO + O_2 \rightarrow 2\ CO_2$). The principal source of CO poisoning is smoke inhalation from house fires, poorly functioning heating systems, and the exhaust from automobile engines (1) (*Note:* The mandate for catalytic converters in automobiles, which converts CO to CO_2, has reduced CO emissions by more than 95%.)

Pathophysiology

Carbon monoxide binds to the heme moieties in hemoglobin (at the same site that binds oxygen) to produce *carboxyhemoglobin* (COHb). The affinity of CO for binding to hemoglobin is over 200 times greater than the affinity of O_2, and CO pressures of only 0.4 mm Hg can fully saturate hemoglobin (2). The effects of COHb on systemic oxygenation are demonstrated by the oxyhemoglobin dissociation curves in Figure 53.1 (1). The curves in this graph show the relationship between oxygen tension (PO_2) and the oxygen content in blood when hemoglobin is normal (upper curve), and when COHb makes up 50% of the hemoglobin molecules (lower curve) (3). The arterial O_2 content (point A) decreases in proportion to the increase in COHb, reflecting the ability of CO to block O_2 binding to hemoglobin. The venous point (point V) on both curves was identified by assuming a normal arteriovenous O_2 content difference (CaO_2 –

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CvO_2 = 5 mL/dL) for both curves. The venous PO_2 (PvO_2) is a close approximation of tissue PO_2, and it is much lower when COHb is present. This provides indirect evidence that CO poisoning impairs tissue oxygenation.

Other Effects

The affinity of CO for heme results in inhibition of other heme-containing proteins, such as myoglobin (which is responsible for O_2 storage in skeletal muscle) and cytochrome oxidase (which is responsible for the oxidative production of ATP in mitochondria) (3). Both effects promote a switch to anaerobic metabolism. CO also activates neutrophils (4), which promotes oxidative tissue injury from reactive oxygen species. (See Chapter 17, and Figure 17.1, for the production of reactive oxygen species during neutrophil activation.) These effects may play an important role in CO poisoning, because *there is a poor correlation between COHb levels and the clinical severity of CO poisoning* (4,5).

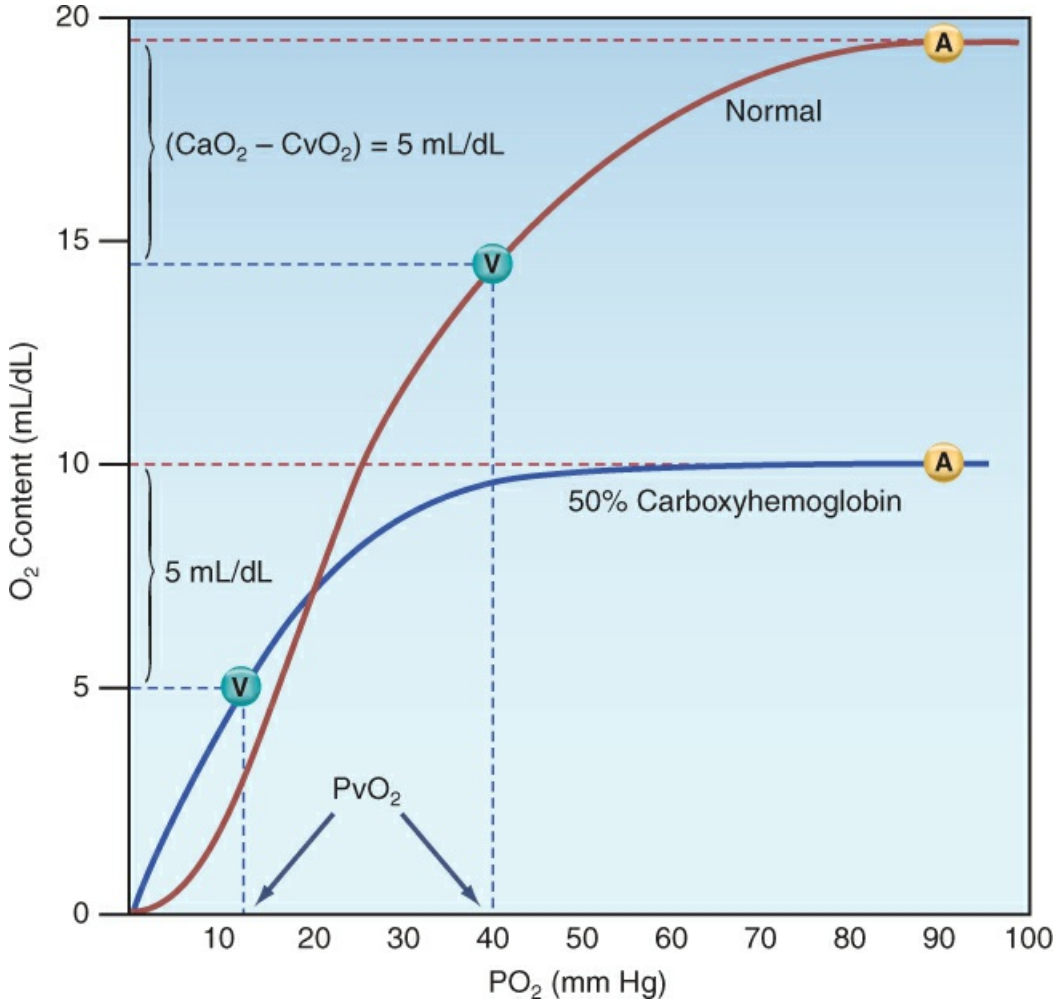


FIGURE 53.1 Influence of 50% carboxyhemoglobin on the oxyhemoglobin dissociation curve (using O_2 content instead of % hemoglobin saturation on the vertical axis). A = arterial blood, V = venous blood. The venous PO_2 (PvO_2) is a reflection of tissue PO, and is decreased by the carbon monoxide effect on hemoglobin. Curves adapted from Reference 1.

Clinical Features

The following statements summarize the clinical manifestations of CO poisoning, which are variable and nonspecific (1).

. Headache (usually frontal) and dizziness are the most common early complaints (1), and may be accompanied by drowsiness, nausea, and vomiting. As CO exposure increases, ataxia appears, along with agitation and changes in mentation (usually confusion and delirium).

. Severe cases of CO poisoning are characterized by obtundation or coma, generalized seizures, respiratory failure, often from the acute respiratory distress syndrome, hemodynamic instability, and lactic acidosis.

. A cherry-red skin color is classically described in CO poisoning (due to COHb being a brighter shade of red than oxyhemoglobin), but this is a rare occurrence (5).

Delayed Neurologic Sequelae

Symptoms of acute CO poisoning can be followed (in one to four weeks) by a neuropsychiatric syndrome that includes mental deterioration (e.g., impaired judgment, memory loss, cognitive defects), and parkinsonism (1,6). The culprit in the delayed neurologic response is unclear, because the correlation with COHb levels has been difficult to establish (1).

Diagnosis

The diagnosis of CO poisoning is not possible based on signs and symptoms alone, and requires evidence of an elevated COHb level in blood.

Carboxyhemoglobin

The measurement of hemoglobin in its different forms (oxygenated and deoxygenated hemoglobin, methemoglobin, and carboxyhemoglobin) is based on the light absorbing properties of each form. This method is known as *oximetry*, and is described in detail in Chapter 7. The following statements summarize the use of oximetry to measure COHb levels in blood.

. *Pulse oximetry is NOT reliable for the detection of COHb*. Pulse oximeters use 2 wavelengths of light to measure oxygenated and deoxygenated hemoglobin in blood. Light absorbance at one of the wavelengths (660 nm) is very similar for oxygenated hemoglobin and COHb (see Figure 7.1), so *COHb is measured as oxygenated hemoglobin (HbO_2) by pulse oximeters, which results in spuriously high readings for HbO_2* (7).

. The measurement of COHb requires an 8-wavelength oximeter (known as a CO-oximeter) that is available in most clinical laboratories. This device measures the relative abundance of all forms of hemoglobin in blood, and the abundance of each form is expressed as a percentage of the total hemoglobin in blood.

COHb levels are negligible (<1%) in healthy nonsmokers, but smokers have COHb levels of 3–5% or even higher (5). Therefore, the threshold for elevated COHb levels is 3–4% for nonsmokers, and 10% for smokers (5).

Management

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The treatment for CO poisoning is inhalation of 100% oxygen. The elimination half-life of COHb is 320 minutes while breathing room air, and 74 minutes while breathing 100% oxygen (5), so *less than 1½ hours of breathing 100% oxygen should reduce COHb levels to normal*. To achieve this, high-flow, heated and humidified nasal O_2, which delivers O_2 at rates up to 40 L/min, seems the best option to ensure 100% O_2 inhalation. (High-flow nasal O_2 is described in Chapter 25, and is depicted in Figure 25.5.)

Hyperbaric Oxygen

Hyperbaric oxygen has been studied for the treatment of CO poisoning since the 1970s, and there is some evidence that it might reduce the risk of delayed neuropsychiatric sequelae (8). However, the sum of the entire experience with hyperbaric O_2 shows no convincing evidence that it is superior to "normobaric" O_2 therapy for CO poisoning (7,9).

Cardiac Concerns

Myocardial injury has been reported in close to 40% of patients with moderate to severe CO poisoning, and is a poor prognostic finding (10). Therefore, an ECG and cardiac biomarkers are recommended for all patients with moderate or severe CO poisoning (7).

Cyanide Poisoning

House fires generate hydrogen cyanide gas as well as carbon monoxide, and empiric treatment for cyanide poisoning is recommended for smoke inhalation victims who are hemodynamically unstable, or have a serum lactate level ≥10 mmol/L (5). This treatment is described in the next section.

CYANIDE POISONING

Cyanide is a lethal toxin with a nefarious history. The Nazis used hydrogen cyanide gas (Zyklon-B) for mass murder in the 1940's, and in 1978, cyanide-laced fruit drinks were used for the mass murder of more than 900 adults and children in Jonestown, Guyana (11). The principal sources of cyanide poisoning include inhalation of hydrogen cyanide gas during domestic fires (12), and oral ingestion of potassium or sodium cyanide, which is converted to hydrogen cyanide by gastric acidity (13). There is also an iatrogenic source of cyanide poisoning from vasodilator infusions with *sodium nitroprusside* (14), which should never be overlooked.

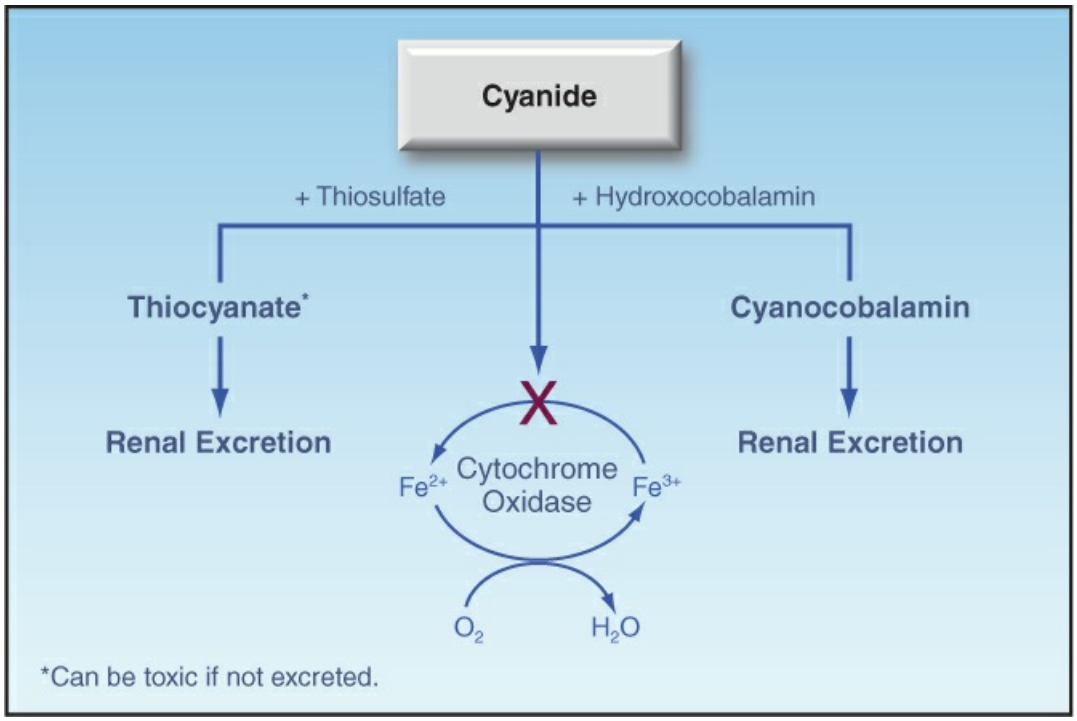


FIGURE 53.2 The actions of cyanide to inhibit cytochrome oxidase, and the clearance of cyanide by hydroxocobalamin and thiocyanate. See text for further explanation.

Pathogenesis

Cyanide ions have a high affinity for metalloproteins, including the oxidized iron (Fe^{3+}) in cytochrome oxidase and methemoglobin. Cytochrome oxidase is the last enzyme system in the electron-transport chain within mitochondria, where the electrons collected during ATP production are used to reduce oxygen to water (see Figure 53.2). Cyanide-induced inhibition of cytochrome oxidase thus halts the process of oxidative energy production in mitochondria. This impedes the uptake of pyruvate into mitochondria, and subsequently increases the production of lactic acid. The accumulation of lactate in plasma produces *a progressive metabolic (lactic) acidosis*, which *is a hallmark of cyanide poisoning*. The defect in cellular energy production is often fatal if not corrected.

Clinical Features

Early signs of cyanide poisoning include agitation, tachycardia, hypertension, and tachypnea, representing the compensatory stage of metabolic acidosis. This often progresses to loss of consciousness, bradycardia, hypotension, and cardiac arrest. Plasma lactate levels are typically very high (>10 mmol/L), and venous blood can look "arterialized" (and have a high PO_2) because of the marked decrease in tissue O_2 utilization (13). Progression is rapid after smoke inhalation, and the time from onset of symptoms to cardiac arrest can be less than 5 minutes (12). Progression can be much slower after the oral ingestion of cyanide, and clinical manifestations can take several minutes or even hours to appear (13).

Diagnosis

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Cyanide poisoning is a clinical diagnosis. Whole blood cyanide levels can be used for documentation, but results are not immediately available, and cyanide antidotes must be given quickly for optimal results. The clinical diagnosis of cyanide poisoning is particularly challenging in victims of smoke inhalation, because many of the clinical features of cyanide poisoning are indistinguishable from carbon monoxide (CO) poisoning. Cyanide poisoning should be suspected if smoke inhalation is associated with hemodynamic instability, or a plasma lactate >10 mmol/L (5). However, there may not be not enough time to wait for a lactate level after smoke inhalation.

Treatment

There are three antidotes for cyanide poisoning, which are included in Table 53.1. Unfortunately, the most effective antidotes must be given intravenously, which can cause dangerous delays in treatment (especially for smoke inhalation victims).

TABLE 53.1 Antidotes for Cyanide Poisoning

Antidote	Dosing Regimens and Comments
Hydroxocobalamin	Dosing: 5 grams infused IV over 15 minutes. Give 10 grams for cardiac arrest. Comment: The antidote of choice for cyanide poisoning. May cause reddish color in urine for a few days.
Sodium Thiosulfate	Dosing: 12.5 grams in 50 mL sterile water, by IV injection. Comment: Used in combination with hydroxocobalamin. Do not use in patients with renal failure.
Amyl Nitrite Inhalant	Dosing: Inhale for 15 sec. then rest for 15 sec, and repeat if needed. Use a new ampule (0.3 mL) every 3 minutes. Comment: Used only for temporary relief when IV access not available. Contraindicated in smoke inhalation.

From References 12, 15, and 18.

Hydroxocobalamin

The antidote of choice for cyanide poisoning is *hydroxocobalamin*, a cobalt-containing precursor of vitamin B_{12} that combines with cyanide to form cyanocobalamin, which is then excreted in the urine (see Figure 53.2). The recommended dose is *5 grams, given as an IV infusion over 15 minutes* (15). A second dose of 5 grams is recommended for life-threatening cases. Hydroxocobalamin is generally considered to be safe, although it can cause a transient rise in blood pressure. It can also produce a reddish color in urine and other body fluids for a few days (16).

Sodium Thiosulfate

Sodium thiosulfate (S_2O_3) converts cyanide to thiocyanate (SCN) by the following *transulfuration reaction*:

$$S_2O_3 + CN \rightarrow SCN + SO_3 \qquad (53.1)$$

Thiocyanate is then cleared by the kidneys. Thiosulfate can be used in combination with hydroxocobalamin (17), and the recommended dose is *12.5 grams IV* (diluted in 50 mL sterile water) (18). However, thiosulfate is not recommended in patients with renal failure because *thiocyanate is a neurotoxin that can accumulate in renal failure and cause an acute psychosis* (14),

Nitrites

Nitrites enhance cyanide clearance by promoting the formation of methemoglobin ($Hb\text{-}Fe^{3+}$), which combines with cyanide to form cyanmethemoglobin: i.e.,

$$Hb\text{-}Fe^{3+} + CN \rightarrow Hb\text{-}Fe^{2+} - CN \qquad (53.2)$$

The hemoglobin-bound cyanide is eventually cleared by reacting with thiosulfate, which should be used in combination with nitrites. Unfortunately, nitrites have undesirable side effects, and *are contraindicated in smoke inhalation* (because methemoglobin promotes tissue hypoxia) (12). The only role for nitrites in cyanide poisoning is the use of inhaled amyl nitrite as a temporary measure when IV access is not available.

Cyanide Antidote Kits

There are two kits available for cyanide poisoning, but neither kit has all the antidotes.

. The Cyanokit® (14) contains 5 grams of hydroxocobalamin as a powder, but has no diluent for IV administration, and has no repeat dose of hydroxocobalamin (14). There is also no thiosulfate or amyl nitrite in this kit.

. The Cyanide Antidote Kit (18) contains 2 vials of sodium thiosulfate for injection: each vial contains 12.5 grams of sodium thiosulfate in 50 mL of sterile water. The kit also contains 12 ampules of 0.3 mL amyl nitrite for inhalation. Each inhalation should be for 15 sec, alternating with 15 sec of rest, and continued as long as needed. A new ampule should be used every 3 minutes.

It certainly would be advantageous if all the cyanide antidotes were available in a single kit.

TOXIC ALCOHOLS

Ethylene glycol and methanol are common components of household, automotive, and industrial products, and they both produce toxic syndromes that are characterized by a metabolic acidosis. They are called *toxic alcohols* (19), but this is a misnomer, because it implies that ethanol is non-toxic.

Ethylene Glycol

Ethylene glycol is the main ingredient in many automotive antifreeze and deicer products. It has a sweet, agreeable taste, which makes it a popular method of attempted suicide.

Pathogenesis

Ethylene glycol is readily absorbed from the GI tract, and 80% of the ingested dose is metabolized in the liver. The metabolism of ethylene glycol involves the formation of a series of acids, with the participation of alcohol dehydrogenase and lactate dehydrogenase enzymes, *ending with the formation of oxalic acid* (see Figure 53.3) (19). Each of the intermediate reactions involves the conversion of NAD to NADH, which promotes the conversion of pyruvate to lactate. As a result, serum lactate levels are also elevated in ethylene glycol poisoning (12). Each of the acid intermediates in ethylene glycol metabolism is a strong acid that readily dissociates, and can contribute to a metabolic acidosis. The oxalic acid also combines with calcium to form insoluble calcium oxalate crystals that precipitate in several tissues, and are particularly prominent in the renal tubules. These crystals are a source of renal tubular injury, and are visible on urine microscopy.

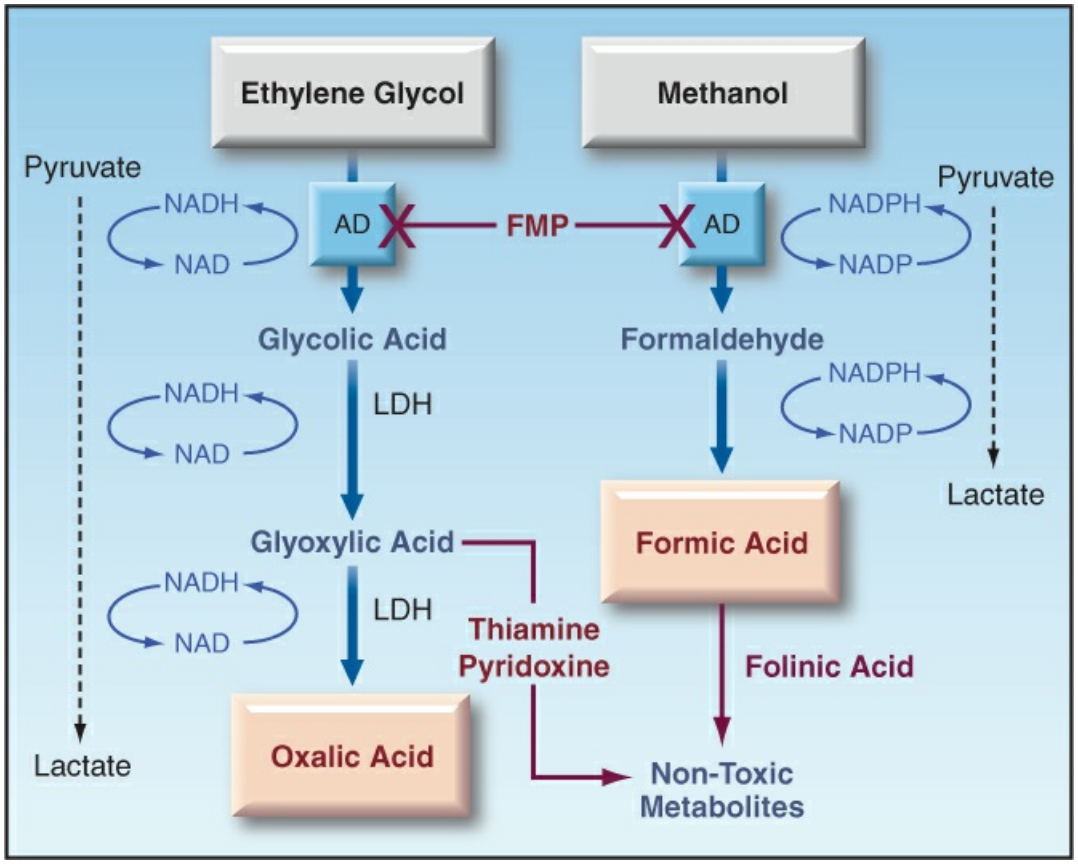


FIGURE 53.3 The metabolism of ethylene glycol and methanol in the liver. AD = alcohol dehydrogenase, LDH = lactate dehydrogenase, FMP = fomepizole.

Clinical Features

Early signs of ethylene glycol intoxication include nausea, vomiting, and apparent inebriation (altered mental status, slurred speech, and ataxia). Because ethylene glycol is odorless, there is no odor of alcohol on the breath. Severe cases are accompanied by depressed consciousness, coma, generalized seizures, renal failure, pulmonary edema, and cardiovascular collapse (19). Renal failure can be a late finding (24 hours after ingestion).

Laboratory studies show a metabolic acidosis with an elevated anion gap (due to glycolic

acid and oxalic acid) and an elevated osmolal gap (See the latter part of Chapter 31 for a description of the osmolal gap.) Serum lactate levels can be elevated, and this can be misleading, as the high anion-gap metabolic acidosis may be mistakenly attributed to the elevated lactate levels.

Plasma assays are available for ethylene glycol and glycolic acid, but neither is readily available, and decisions about acute management are based on clinical judgment. However, plasma levels of ethylene glycol and glycolic acid should be obtained for confirmatory purposes. Plasma levels of ethylene glycol that are above 20 mg/dL are considered toxic (19), but plasma glycolic acid levels are more predictive of the severity of disease, and levels above 8 mmol/L have been proposed as an indication for hemodialysis (20).

CRYSTALLURIA: Calcium oxalate crystals can be visualized in the urine in about 50% of cases of ethylene glycol poisoning (21). The presence of calcium oxalate crystals is not specific for ethylene glycol poisoning, but the shape of the crystals is more specific; i.e., thin, monohydrate crystals, like the ones shown in Figure 53.4, are more characteristic of ethylene glycol poisoning than the box-shaped dihydrate crystals (21). Most hospital laboratories do not routinely inspect urine for crystals, so make sure to request a search for crystals when a urine sample is sent to the laboratory.

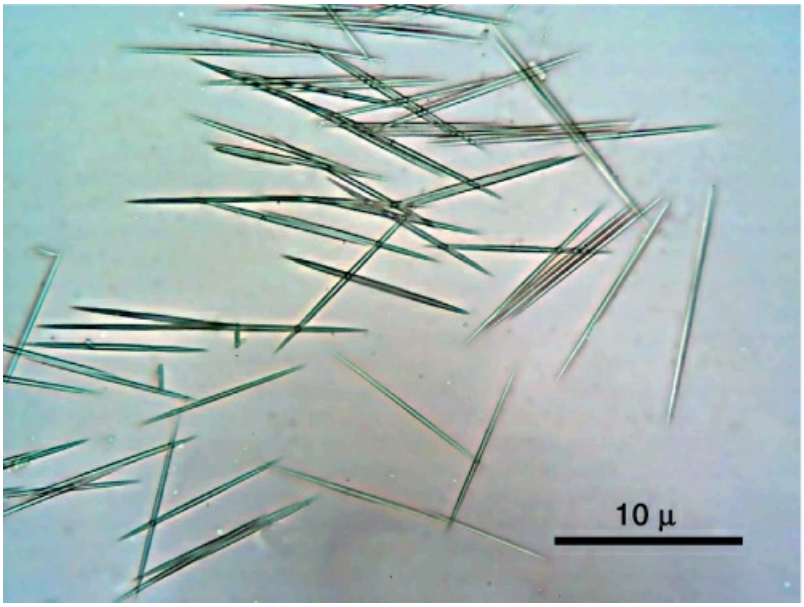


FIGURE 53.4 Microscopic appearance of calcium oxalate monohydrate crystals. The presence of these thin, needle-shaped crystals in the urine is highly suggestive of ethylene glycol poisoning.

Treatment

The management of ethylene glycol poisoning involves measures to alter the metabolism of ethylene glycol, along with hemodialysis, if necessary.

FOMEPIZOLE: Fomepizole inhibits alcohol dehydrogenase, the enzyme involved in the initial step of ethylene glycol metabolism (see Figure 53.3). The recommended dosing regimen for both

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ethylene glycol and methanol poisoning is shown in Table 53.2 (19). The best results are obtained if therapy begins within 4 hours of ingestion. Fomepizole should be continued until the toxic alcohol levels in plasma have fallen to <20 mg/dL, and the metabolic acidosis has resolved.

TABLE 53.2 Dosing Regimen for Fomepizole

1. Start with an IV loading dose of 15 mg/kg.
2. Follow with a dose of 10 mg/kg IV every 12 hrs for 4 doses.
3. If the toxic alcohol level is >20 mg/dL after 4 doses, increase the dose to 15 mg/kg IV every 12 hours, and continue until the following end-points are reached:
   a. the toxic alcohol level is <20 mg/dL,
   b. the plasma pH is normal,
   c. the patient is asymptomatic.
4. If more than one hemodialysis session is required, change the dose to 15 mg/kg IV every 4 hrs, until dialysis is no longer necessary.

From Reference 19.

HEMODIALYSIS: The clearance of ethylene glycol and all its metabolites is enhanced by hemodialysis. The indications for immediate hemodialysis include severe acidemia (pH <7.1), and evidence of significant end-organ damage (e.g., coma, seizures, and renal insufficiency) (19). Multiple courses of hemodialysis may be necessary, and fomepizole dosing should be adjusted, as indicated in Table 53.2.

ADJUNCTS: Thiamine (100 mg IV daily) and pyridoxine (100 mg IV daily) are recommended to divert glyoxylic acid to the formation of non-toxic metabolites (see Figure 53.3).

Methanol

Methanol (also known as *wood alcohol* because it was first distilled from wood) is a common ingredient in shellac, varnish, paint remover or thinner, windshield washer fluid, and solid cooking fuel (e.g., Sterno® products).

Pathogenesis

Like ethylene glycol, methanol is readily absorbed from the upper GI tract, and is metabolized by alcohol dehydrogenase in the liver. The *principal metabolite is formic acid*, a strong acid that readily dissociates and produces a metabolic acidosis with a high anion gap (19). Formic acid is also a mitochondrial toxin that inhibits cytochrome oxidase and blocks oxidative energy production. Tissues that are particularly susceptible to damage are the retina, optic nerve, and basal ganglia. In one study, hypodense lesions in the basal ganglia were present on CT scan in 64% of patients with methanol poisoning (22). Methanol metabolism promotes the conversion of pyruvate to lactate in the same manner described for ethylene glycol metabolism, and lactate production is increased further by the toxic effects of formic acid.

Clinical Features

Early manifestations (within 6 hours of ingestion) include signs of apparent inebriation without

the odor of ethanol (as in ethylene glycol intoxication). Later signs (6–24 hours after ingestion) include visual disturbances (e.g., scotoma, blurred vision, complete blindness), depressed consciousness, coma, and generalized seizures (21). Examination of the retina can reveal papilledema and generalized retinal edema. The visual disturbances are characteristic of methanol poisoning, and are not a feature of ethylene glycol poisoning.

Laboratory studies show a metabolic acidosis, high anion gap, and high osmolal gap, similar to ethylene glycol poisoning. However, there is no crystalluria in methanol poisoning. A plasma assay for methanol is available, and a level above 20 mg/dL is considered toxic. However, the results of the plasma assay are not immediately available, and are not used in the decision to initiate therapy.

Treatment

Treatment for methanol poisoning is the same as described for ethylene glycol poisoning, except for the following: (a) visual impairment is an indication for dialysis in methanol poisoning and (b) folinic acid is used as adjunctive therapy in methanol poisoning, instead of thiamine and pyridoxine.

FOLINIC ACID: Folinic acid (leucovorin) can convert formic acid to non-toxic metabolites. The recommended dose is 1 mg/kg IV, up to 50 mg, at 4-hour intervals (21). Folic acid should be used if folinic acid is unavailable.

A FINAL WORD

The following points in this chapter deserve emphasis:

. Carbon monoxide poisoning can be missed if you rely on pulse oximetry readings to detect O_2 desaturation. Detection of elevated carboxyhemoglobin levels requires an 8-wavelength CO-oximeter, which is in most clinical laboratories.

. A victim of smoke inhalation who has a severe lactic acidosis or hemodynamic instability should be treated empirically for cyanide poisoning.

. Methanol and ethylene glycol poisoning should be considered in any patient who presents with a high-anion-gap metabolic acidosis of unclear etiology.

. If ethylene glycol poisoning is a consideration, inspection of the urine for calcium oxalate monohydrate crystals can provide confirmatory evidence.

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Cyanide

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Section XVII

APPENDICES

When you're through learning, you're through.

Will Rogers

Appendix 1

Units and Conversions

The units of measurements in the medical sciences are taken from the metric system (centimeter, gram, second) and the Anglo-Saxon system (foot, pound, second). The metric units were introduced during the French Revolution and were revised in 1960. The revised units are called Système Internationale (SI) units and are currently the worldwide standard.

PART 1 Units of Measurement in the Système Internationale

Parameters	Dimensions	Basic SI Unit (Symbol)	Equivalencies
Length	L	Meter (m)	1 inch = 2.54 cm
Area	L^2	Square meter (m^2)	1 square centimeter (cm^2) = 104 m^2
Volume	L^3	Cubic meter (m^3)	1 liter (L) = 0.001 m^3 1 milliliter (mL) =1 cubic centimeter (cm^3)
Mass	M	Kilogram (kg)	1 pound (lb) = 453.5 g 1 kg = 2.2 lbs
Density	M/L^3	Kilogram per cubic meter (kg/m^3)	1 kg/m^3 = 0.001 kg/dm^3 Density of water = 1.0 kg/dm^3 Density of mercury = 13.6 kg/dm^3
Velocity	L/T	Meters per second (m/sec)	1 mile per hour (mph) = 0.4 m/sec
Acceleration	L/T^2	Meters per second squared (m/sec^2)	1 ft/sec^2 = 0.03 m/sec^2
Force	M x (L/T^2)	Newton (N) = kg × (m/sec^2)	1 dyne = 10^{-5} N

PART 2 Units of Measurement in the Système Internationale

Parameters	Dimensions	Basic SI Unit (Symbol)	Equivalencies
Pressure	M × (L/T^2)	Pascal (Pa) = N/m^2	1 kPa = 7.5 mm Hg = 10.2 cm H_2O 1 mm Hg = 1 × 10^{-9} torr (See conversion table for kPa and cm H_2O).

Heat	M × (L/T^2) × L	Joule (J) = N x M	1 kilocalorie (kcal) = 4,184 J
Temperature	None	Kelvin (K)	0° C = –273 K (See conversion table for °C and °F)
Viscosity	M, 1/L, 1/T	Newton × second per square meter (N • sec/m^2)	Centipoise (cP) = 10^{-3} N • sec/m^2
Amount of a substance	N	Mole (mol) = molecular weight in grams	Equivalent (Eq) = mol × valence

PART 1 Converting Units of Solute Concentration

1. For ions that exist freely in an aqueous solution, the concentration is expressed as milliequivalents per liter (mEq/L). To convert to millimoles per liter (mmol/L):

$$\frac{\text{mEq/L}}{\text{valence}} = \text{mmol/L}$$

   a. For a univalent ion like potassium (K^+), the concentration in mmol/L is the same as the concentration in mEq/L.

   b. For a divalent ion like magnesium (Mg^{++}), the concentration in mmol/L is one-half the concentration in mEq/L.

2. For ions that are partially bound or complexed to other molecules (e.g., plasma Ca^{++}), the concentration is usually expressed as milligrams per deciliter (mg/dL). To convert to mEq/L:

$$\frac{\text{mg/dL} \times 10}{\text{mol wt}} \times \text{valence} = \text{mEq/L}$$

   where mol wt is molecular weight, and the factor 10 is used to convert deciliters (100 mL) to liters.

   EXAMPLE: Ca^{++} has a molecular weight of 40 and a valence of 2, so a plasma Ca^{++} concentration of 8 mg/dL is equivalent to:

$$(8 \times 10/40) \times 2 = 4 \text{ mEq/L}$$

PART 2 Converting Units of Solute Concentration

3. The concentration of uncharged molecules (e.g., glucose) is also expressed as milligrams per deciliter (mg/dL). To convert to (mmol/L):

$$\frac{\text{mg/dL} \times 10}{\text{mol wt}} = \text{mmol/L}$$

   EXAMPLE: Glucose has a molecular weight of 180, so a plasma glucose concentration of 90 mg/dL is equivalent to: (90 × 10/180) = 5 mmol/L.

4. The concentration of solutes can also be expressed in terms of osmotic pressure, which determines the distribution of water in different fluid compartments. Osmotic activity in aqueous solutions (called osmolality) is expressed as milliosmoles per kg water (mosm/kg H_2O or mosm/kg).

   The following formulas can be used to express the osmolality of solute concentrations (n is the number of nondissociable particles per molecule).

$$\text{mmol/L} \times n = \text{mosm/kg}$$

$$\frac{\text{mEq/L}}{\text{valence}} \times n = \text{mosm/kg}$$

$$\frac{\text{mg/dL} \times 10}{\text{mol wt}} \times n = \text{mosm/kg}$$

EXAMPLE:

a. A plasma Na^+ concentration of 140 mEq/L has the following osmolality:

$$\frac{140}{1} \times 1 = 140 \text{ mosm/kg}$$

b. A plasma glucose concentration of 90 mg/dL has the following osmolality:

$$\frac{90 \times 10}{180} \times 1 = 5 \text{ mosm/kg}$$

The sodium in plasma has a much greater osmotic activity than the glucose in plasma because osmotic activity is determined by the number of particles in solution, and is independent of the size of the particles (i.e., one sodium ion has the same osmotic activity as one glucose molecule).

Apothecary and Household Conversions

Apothecary	Household
1 grain = 60 mg 1 ounce = 30 mg 1 fluid ounce = 30 mL 1 pint = 500 mL 1 quart = 947 mL	1 teaspoonful = 5 mL 1 tablespoonful = 15 mL 1 wineglassful = 60 mL 1 teacupful = 120 mL

French Sizes

French Size	Outside Diameter*	
	Inches	mm
1	0.01	0.3
4	0.05	1.3
8	0.10	2.6
10	0.13	3.3
12	0.16	4.0
14	0.18	4.6
16	0.21	5.3
18	0.23	6.0

20	0.26	6.6
22	0.28	7.3
24	0.31	8.0
26	0.34	8.6
28	0.36	9.3
30	0.39	10.0
32	0.41	10.6
34	0.44	11.3
36	0.47	12.0
38	0.50	12.6

*Diameters can vary with manufacturers. However, a useful rule of thumb is OD (mm) × 3 = French size.

Gauge Sizes

Gauge Size	Outside Diameter*	
	Inches	mm
26	0.018	0.45
25	0.020	0.50
24	0.022	0.56
23	0.024	0.61
22	0.028	0.71
21	0.032	0.81
20	0.036	0.91
19	0.040	1.02
18	0.048	1.22
16	0.040	1.62
14	0.080	2.03
12	0.104	2.64

*Diameters can vary with manufacturers.

Temperature Conversions

°C	°F	°C	°F
41	105.8	35	95

A L G r a w a n y

40	104	34	93.2
39	102.2	33	91.4
38	100.4	32	89.6
37	98.6	31	87.8
36	96.8	30	86
°F = (9/5 °C) + 32		°C = 5/9 (°F – 32)	

Pressure Conversions

mm Hg	kPa	mm Hg	kPa	mm Hg	kPa
41	5.45	61	8.11	81	10.77
42	5.59	62	8.25	82	10.91
43	5.72	63	8.38	83	11.04
44	5.85	64	8.51	84	11.17
45	5.99	64	8.51	85	11.31
46	6.12	66	8.78	86	11.44
47	6.25	67	8.91	87	11.57
48	6.38	68	9.04	88	11.70
49	6.52	68	9.04	89	11.84
50	6.65	70	9.31	90	11.97
51	6.78	71	9.44	91	12.10
52	6.92	72	9.58	92	12.24
53	7.05	73	9.71	93	12.37
54	7.18	74	9.84	94	12.50
55	7.32	75	9.98	95	12.64
56	7.45	76	10.11	96	12.77
57	7.58	76	10.11	97	12.90

Kilopascal (kPa) = 0.133 × mm Hg; mm Hg = 7.5 × kPa.

Appendix 2

Selected Reference Ranges

Radiation Dose in Common Radiographic Procedures

Procedure	Average Effective Dose[1]	Range
Chest x-ray (1 view) Chest x-ray (2 views) Abdomen	0.02 mSv 0.1 mSv 0.7 mSv	0.007–0.05 0.05–0.24 0.04–1.1
Head CT Chest CT Abdominal CT	2 mSv 7 mSv 8 mSv	0.9–4.0 4–18 3.5–25
Chest CT Angiography Coronary Angiography	15 mSv 16 mSv	13–40 5–32
Annual Radiation Limit[2]	50 mSv	

From Mettler FA Jr, et al. Effective doses in radiology and diagnostic nuclear medicine: A catalogue. Radiology 2008; 248:254–263.

[1] The effective dose (in millisieverts or mSv) expresses the radiation dose to all organs in the body. [2] From the Environmental Protection Agency.

Body Mass Index

A L G r a w a n y

Body Mass Index

WEIGHT	lbs	100	105	110	115	120	125	130	135	140	146	150	155	160	165	170	175	180	185	190	195	200	205	210	215
	kg	45.5	47.7	50.0	52.3	54.5	56.8	59.1	61.4	63.6	65.9	68.2	70.5	72.7	75.0	77.3	79.5	81.8	84.1	86.4	88.6	90.9	93.2	95.5	97.7
HEIGHT in	cm	Underweight					Healthy					Overweight					Obese				Extremely Obese				
5.'0"	152.4	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42
5.'1"	154.9	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	36	37	38	39	40
5.'2"	157.4	18	19	20	21	22	22	23	24	25	26	27	28	29	30	31	32	33	33	34	35	36	37	38	39
5.'3"	160.0	17	18	19	20	21	22	23	24	24	25	26	27	28	29	30	31	32	32	33	34	35	36	37	38
5.'4"	162.5	17	18	18	19	20	21	22	23	24	24	25	26	27	28	29	30	31	31	32	33	34	35	36	37
5.'5"	165.1	16	17	18	19	20	20	21	22	23	24	25	25	26	27	28	29	30	30	31	32	33	34	35	35
5.'6"	167.6	16	17	17	18	19	20	21	21	22	23	24	25	25	26	27	28	29	29	30	31	32	33	34	34
5.'7"	170.1	15	16	17	18	18	19	20	21	22	22	23	24	25	25	26	27	28	29	29	30	31	32	33	33
5.'8	172.7	15	16	16	17	18	19	19	20	21	22	22	23	24	25	25	26	27	28	28	29	30	31	32	32
5.'9"	175.2	14	15	16	17	17	18	19	20	20	21	22	22	23	24	25	25	26	27	28	28	29	30	31	31
5.'10"	177.8	14	15	15	16	17	18	18	19	20	20	21	22	23	23	24	25	25	26	27	28	28	29	30	30
5.'11"	180.3	14	14	15	16	16	17	18	18	19	20	21	21	22	23	23	24	25	25	26	27	28	28	29	30
6.'0"	182.8	13	14	14	15	16	17	17	18	19	19	20	21	21	22	23	23	24	25	25	26	27	27	28	29
6.'1"	185.4	13	13	14	15	15	16	17	17	18	19	19	20	21	21	22	23	23	24	25	25	26	27	27	28
6.'2"	187.9	12	13	14	14	15	16	16	17	18	18	19	19	20	21	21	22	23	23	24	25	25	26	27	27
6.'3"	190.5	12	13	13	14	15	15	16	16	17	18	18	19	20	20	21	21	22	23	23	24	25	25	26	26
6.'4"	193.0	12	12	13	14	14	15	15	16	17	17	18	18	19	20	20	21	22	22	23	23	24	25	25	26

Desirable Weights (in lbs.) for Adults*

Desirable Weights (in lbs.) for Adults*				
Height		Males		
Feet	Inches	Small Frame	Medium Frame	Large Frame
5	2	128–134	131–141	138–150
5	3	130–136	133–143	140–153
5	4	132–138	135–145	142–156
5	5	134–140	137–148	144–160
5	6	136–142	139–151	146–164
5	7	138–145	142–154	149–168
5	8	140–148	145–157	152–172
5	9	142–151	148–160	155–176
5	10	144–154	151–163	158–180
5	11	146–157	154–166	161–184
6	0	149–160	157–170	164–188
6	1	152–164	160–174	168–192
6	2	155–168	164–178	172–197
6	3	158–172	167–182	172–202
6	4	162–176	171–187	181–207

A L G r a w a n y

Height		Females		
Feet	Inches	Small Frame	Medium Frame	Large Frame
4	10	102–111	109–121	112–131
4	11	103–113	111–123	120–134
5	0	104–115	113–126	122–137
5	1	106–118	115–129	125–140
5	2	108–121	118–132	128–143
5	3	111–124	121–135	131–147
5	4	114–127	124–138	134–151
5	5	117–130	127–141	137–155
5	6	120–133	130–144	140–159
5	7	123–136	133–147	143–163
5	8	126–139	136–150	146–167
5	9	129–142	139–153	149–170
5	10	132–145	142–156	152–173
5	11	135–148	145–159	155–176
6	1	138–151	148–162	158–179

*Unclothed weights associated with the longest life expectancies. From the statistics bureau of the Metropolitan Life Insurance Company, 1983.

Peak Expiratory Flow Rates for Healthy Males

Age (yr)	Ht:	L/min			
		60″	65″	70″	75″
20		602	649	693	740
25		590	636	679	725
30		577	622	664	710
35		565	609	651	695
40		552	596	636	680
45		540	583	622	665
50		527	569	607	649
55		515	556	593	634
60		502	542	578	618
65		490	529	564	603
70		477	515	550	587

Peak Flow (L/min) = [3.95–(0.0151 × Age)] × Ht (cm).
Regression equation from Leiner GC, et al. Am Rev Respir Dis 1963; 88:646.

Peak Expiratory Flow Rates for Healthy Females

A L G r a w a n y

Age (yr)	Ht:	L/min 60″	65″	70″	75″
20		309	423	460	496
25		385	418	454	490
30		380	413	448	483
35		375	408	442	476
40		370	402	436	470
45		365	397	430	464
50		360	391	424	457
55		355	386	418	451
60		350	380	412	445
65		345	375	406	439
70		340	369	400	432

Peak Flow (L/min) = [2.93 –(0.0072 × Age)] × Ht (cm).

Regression equation from Leiner GC, et al. Am Rev Respir Dis 1963; 88:646.

Estimated Blood Volumes in Adults

	mL/kg Male	Female
Blood	66	60
Red Cells	26	24
Plasma	40	36

From AABB Technical Manual, 10th ed. Arlington, VA: American Association of Blood Banks, 1990.

The Confusion Assessment Method for the ICU (CAM-ICU)

The Confusion Assessment Method for the ICU (CAM-ICU)

	Yes	No
I: Acute Onset or Fluctuating Course: A. Is there evidence of an acute change in metal status? OR B. Did the abnormal behavior fluctuate during the past 24 hrs?	☐	☐
II: Inattention: Did the patient have difficulty focusing attention? (The patient can be shown 5 simple items as asked to recall the items in succession.)	☐	☐
III: Disorganized Thinking: Is there evidence of disorganized or incoherent thinking as evidenced by incorrect answers to 3 or more of the following questions? 1. Will a stone float in water? 2. Are there fish in the sea? 3. Does one pound weigh more than 2 pounds? 4. Can you use a hammer to pound a nail? AND is there difficulty following these commands? 1. Hold up this many fingers. (The examiner holds two fingers in front of the patient.) 2. Now do the same thing with the other hand (without holding the two fingers in front of the patient).	☐	☐
IV: Altered Level of Consciousness: Is the patient's level of consciousness anything other than alert, such as being hyperalert or lethargic?	☐	☐

A Yes answer to I and II, and either III or IV = Delirium.

From Ely EW, et al. JAMA 2001; 286:2703.

A L G r a w a n y

Appendix 3

The Body Fantastic

We cannot fathom the marvelous complexity of an organic being.

Charles Darwin

The real problem in medicine is not the cost or availability of healthcare—it is the task of fixing a machine that has about 100 trillion interconnecting parts (the estimated number of cells in the human body), none of which are visible to the naked eye. The following are some facts and factoids that highlight the complexity of the machine we are tasked to fix. This information has no direct application to the practice of medicine, and is meant only as an homage to the human body.

Our Vascular Superhighway

If all the blood vessels in the human body were placed end-to-end, they would extend about 100,000 kilometers (62,000 miles) (1), which is about 2½ times the circumference of Earth.

1. Vogel S. *Vital Circuits. On Pumps, Pipes, and the Workings of Circulatory Systems*. Oxford: Oxford University Press, 1992:16.

The Reach of Our Airways

The combined length of all the airways in the lungs is estimated at 2,500 kilometers (1,553 miles) (1), which is equivalent to the distance between London and Moscow (1,554 miles).

1. West G. *Scales. The Universal Laws of Life, Growth, and Death in Organisms, Cities, and Companies*. New York: Penguin Books, 2017:100.

The Ultimate Packing Job

The human genome is roughly 2 meters in length, yet it is packed into the cell nucleus, which is only 6 microns in diameter. This is analogous to taking a string that runs up and down the length of Manhattan (roughly 14 miles) and packing it in a tennis ball (diameter 2.6 inches).

The Complexity of the Human Genome

The DNA molecule encodes information using the pairing of four nucleotides: adenine, thymine,

cytosine, and guanine. The human genome contains about 3 billion nucleotide pairings, so the total possible sequences of nucleotide pairings is $4^{3,000,000,000}$. In comparison, the total number of electrons in the universe is estimated at 10^{80}, or 4^{133} (1).

1. Roston E. *The Carbon Age*. New York: Walker & Co, 2008:38.

The Amazing DNA Polymerase Enzyme

The human genome contains 3.2 billion nucleotide pairs, yet each replication of DNA is carried out with remarkable accuracy; i.e., a mistake occurs only once for every 10 billion nucleotides copied (1). The source of this fidelity is the DNA polymerase enzyme, which catalyzes the addition of each nucleotide to the DNA molecule, but checks to see that the nucleotide to be added is the correct one before adding it to the molecule. This means that, for every replication of the DNA molecule, the DNA polymerase enzyme "proofreads" each of the 3.2 billion nucleotide pairs to ensure accuracy of replication. This allows us to reliably transfer genetic information to our offspring.

1. DNA Replication, Repair, and Recombination. In: Alberts B, Johnson A, Lewis J, et al, eds. *Molecular Biology of the Cell*, 6th ed. New York: Garland Science, 2015:237−298.

The Celestial Reach of Our Genome

If all the DNA strands in the body were placed end-to-end, they would stretch an estimated 10 billion miles (1), which is almost three times the distance from the Sun to Pluto (3.7 billion miles).

1. Pollack R. *Signs of Life: The Language and Meanings of DNA*. London: Viking Press, 1994:19.

Human Life and an 80-watt Light Bulb

The daily energy requirement of an adult at rest is about 25 kcal/kg, which is 2,000 kcal for an 80 kg (176 lb) person. This corresponds to a hourly energy requirement of 83 kcal/hr, which is equivalent to 80 watts (1 kcal/hr=1.16 watts). Thus, the energy needed to support human life is about the same as the energy needed to power an 80-watt light bulb.

The Molecular Burden of Staying Alive

The energy used by the human body to conduct the myriad processes involved in living is derived from the production and breakdown of adenosine triphosphate (ATP). The daily production of ATP is estimated at 2×10^{26} molecules per day, which corresponds to a mass of about 80 kg (176 lb) (1). This is close to the average weight of an adult in North America (177.5 lbs) (2). This means that, each and every day, the adult human body cycles a molecular mass equivalent to its own body weight in order to stay alive.

1. West G. *Scaling*. New York: Penguin Books, 2018:100.
2. Walpole SC, Prieto-Merino D, Edwards P, et al. The weight of nations: an estimation of adult human biomass. BMC Public Health 2012; 12:439−445.

Our Erythrocyte Burden

The number of circulating red blood cells (RBCs) is normally about 5.4×10^{12}/L (5.4 trillion/L) in adult males (1). Assuming a normal blood volume of 5 liters, the total number of circulating

RBCs is 27 trillion. About 1% of circulating RBCs are replaced daily, (2) which would be 1% of 27 trillion, or 270 billion RBCs replaced daily (which is equivalent to 3.1 million RBCs/sec). Thus, to maintain the pool of circulating RBCs, the human body must manufacture 3.1 million RBCs every second.

1. Walker RH, ed. *American Association of Blood Banks Technical Manual*, 10th ed, Arlington, VA: American Association of Blood Banks, 1990.
2. Hillman RS, Finch CA. *Red Cell Manual*, 6th ed. Philadelphia: F.A. Davis Co, 1992.

The Incomprehensible Heart Beat

Every heart beat has 6 variables to consider: the part of the cardiac cycle (systole and diastole), the side of the heart (right and left side) and the ventilatory cycle (inhalation and exhalation). Considering that humans can process only 4 independent variables at one time (1), this means a single heart beat is incomprehensible at any point in time.

1. Halford GS, Baker R, McCreddon JE, Bain JD. How many variables can humans process? Psychol Sci 2005; 16:70−76.

Human Molting

The surface of the skin is composed entirely of dead cells, which are regularly shed and replaced. The average rate of shedding is estimated at 25,000 cells per minute, which means that each day, we shed about 36 million skin cells (1). Our skin regenerates about once every 30 days (2), which means that we generate a new skin about 12 times each year.

1. Bryson B. *The Body: A Guide for Occupants*. New York: Anchor Books, 2021:12.
2. Epidermis and its Renewal by Stem Cells. In: Alberts B, Johnson A, Lewis J, et al, eds. *Molecular Biology of the Cell*, 4th ed. New York: Garland Science, 2002.

Counting the Connections in the Cerebral Cortex

The number of synaptic connections in the cerebral cortex is estimated at one million billion, or 10^{15} (1). If you were to count to this number at a rate of one number per second (one mississippi, two mississippi, etc.), it would take about 32 million years to complete the task.

1. Edelman G. *Bright and Brilliant Fire. On the Matters of the Mind*. New York: Basic Books, 1992.

The Storage Capacity of the Brain

The popular claim that we use only a small fraction of our brain's capacity is supported by the report of a Japanese mnemonist (an individual who is skilled at retrieving information from memory) named Akira Hiraguchi, who memorized the sequence of numbers in Pi to 83,431 digits. It took him 16 hours and 28 minutes to recite all the numbers (1).

1. Foer J. *Moonwalking with Einstein. The Art and Science of Remembering Everything.* London: Penguin Books, 2011.

Recommendation

If you enjoy the kind of numerical approach to the human body used in this Appendix, you might also enjoy the following text: Milo R, Phillips R. *Cell Biology by the Numbers*. New York: Garland Science, Taylor & Francis Group, 2016.